

**ADMINISTRATIVE INFORMATION****FMI No: 25390****FMI Title:** Performix Pro VCT 100 (Hercules) Tube Window Screw Replacement

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☒ Mandatory Safety	GEMS - AM:	Issue Date:
☐ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

☒ Staged FMI	Customer List provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
☒	Parts and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
☐	Parts Ordering: GEMS - Americas, Contact your Region logistics coordinator GEMS - Europe, Order FMI kits from GPO/EPL Buc GEMS - Asia, Contact your region logistics coordinator
☐	No Parts are required to complete this FMI.
☐ Batch with	
	To minimize costs, combine implementation of this FMI with the FMIs listed for all customers where the effectivity is the same (Please see your customer lists).
☐ Non-Staged	No Customer List is provided. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center 800-321-7937 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:

If there are consignment tubes onsite, please contact your process leader or FMI Administrator to order additional FMI kits—one kit for each tube—and complete procedures on all tubes.

Unit Tracked: Refer to Effectivity list on page 2.**FMI Instructions Part No.:** 5193259-100, Rev 2**Do NOT leave unsecured on site**

FMI Completion Certification Report

Mail To

GEMS - Americas

G.E. Medical Systems
Attn: FMI Admin. W-523
P.O. Box 414
Milwaukee, WI 53201

GEMS - Europe

GEMS - Europe
FMI Admin. 322
BP34
F78533 Buc Cedex, France

Send to your
Country FMI or Service
Administrator

FMI No.: 25390

FMI Title: Performix Pro VCT 100 (Hercules) Tube Window Screw Replacement

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Location notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer
Comments

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Revision History

Revision	Date	Reason for Change
1	27-OCT-2006	Initial release
2	10-JAN-2007	Add Exclusion definition for previous design.

Effectivity

This FMI applies to all LightSpeed 7.x and 5.x systems with the Performix Pro VCT 100 (Hercules) tube. The products and model numbers affected are:

Product Line	System	Gantry Model
LightSpeed 7.x	VCT 64	5124069
	Discovery VCT (DVCT)	5124069
	VCT 32/Select	5126093
LightSpeed 5.x	LightSpeed Pro 16 (HP100/80)	2357739
		2357739-3*
		2357739-4
		2357739-6
		2357739-7
		2357739-8
	H-Power RT/Extra (16 slice)	2374681-6

*Model No. 2357739-3 only affects LightSpeed 5.x (HP 80) systems with Performix Pro VCT 100 tubes with Cat# D3194T.

Purpose

A few instances of X-ray tube oil leakage involving GE CT scanners have been discovered in the field. The leakage has been traced to the screws mounting the port window to the X-ray tube. Although the problem has been found in only a small number of the CT systems, we will be inspecting and servicing all systems using the Performix Pro VCT 100 tube. Oil on the port window could lead to artifacts on the images produced by the system due to a change in beam filtration.

This addresses the following PSRs:

- ***D3194T Hercules Window Oil Leak*** (13065885)
- ***D3194T Hercules Oil Leak - Image Artifact*** (13085641)
- ***D3194T Hercules - oil bubble in port*** (13086220)
- ***D3194T Hercules Image artifact -Window leak*** (13086187)

Prerequisite FMIs

None.

Related FMIs

None.

Furnished Materials

FMI Kit 5193259 consists of the following materials:

Quantity	Description	Part #
1	FMI document	5193259-100
13	Self sealing M4 X 8 long screws with Silicone O-ring	2292492-2
1	Performix Pro VCT 100 Copper Window Filter.	5126841
14	Phillips flat head machine screws M2 x 6 DIN 965A, Type H, BN388 zinc plated steel property class 4.8	5111548
1	"Window Screw FMI complete" label	5193715
1	Empty 500ml alcohol bottle with spray nozzle	5193711
4	Nitrile Gloves	2170756-4
1	Sharpie Marker	2168609

Tool Requirements

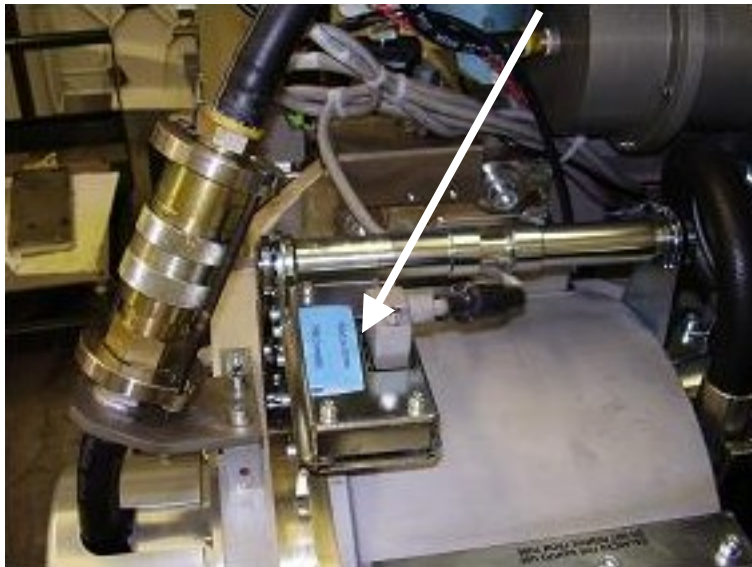
Quantity	Description	Part #
1	Small Phillips screwdriver	Field supplied
1	Medium size flat screw driver	Field supplied
1	Torque Wrench capable of 3.0 N-m (26.6 in-lb)	Field supplied
1	Nitrile Gloves	Field supplied
1	Bottle of alcohol (isopropyl or ethanol) for cleaning parts	Field supplied
1	Can of compressed air	Field supplied
1	Tube Hoist	Field supplied

Personnel Requirements

Resource	1 FE
Labor	8 hrs
Travel	1 hr

Before You Begin

1. If there are consignment tubes onsite, please contact your process leader or FMI Administrator to order additional FMI kits—one kit for each tube—and complete procedures on all tubes.
2. Inspect the Smart ID board cover for the “Window Screw FMI Complete” sticker. If sticker is on tube then FMI has already been completed and the dispatch can be closed as code 1, Modification Complete.



3. Check for Old Casing Design BEFORE tube Removal
An extremely limited number of tubes currently in the field may have an older Casing Window design. Those tubes (described below) with the older Casing Window design do not require the screw replacement. The following instructions will describe the process for determining the need for screw replacement.

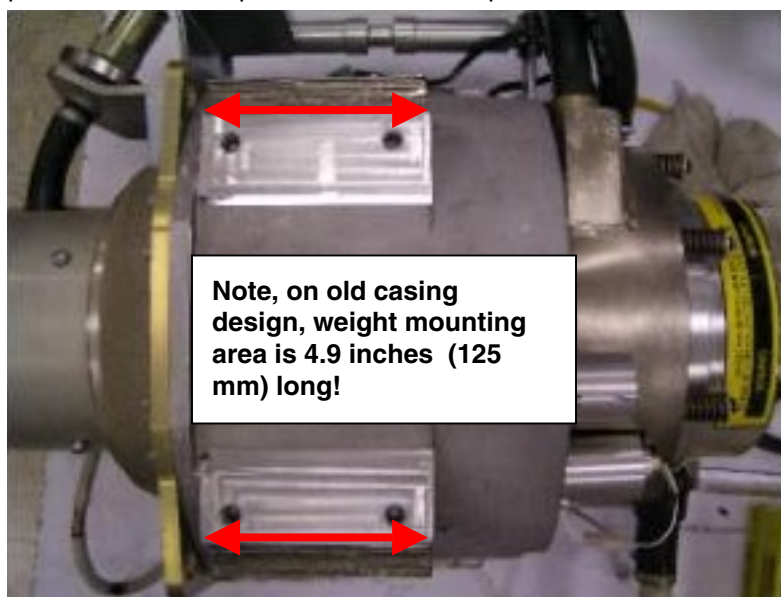
Check Insert and Casing serial numbers. If the insert number is greater than 115893GI0 and the casing number is greater than 12911GM4, **the screws need to be replaced, go to Screw Replacement Procedure on page 7.** If either number is less, the tube needs visual inspection. **Until visual inspection verifies casing design, assume screws need to be replaced.**

Visually inspect / measure casing. The old casing can be identified before removal from the system by inspecting the weight mounting area on the center section of the casing.

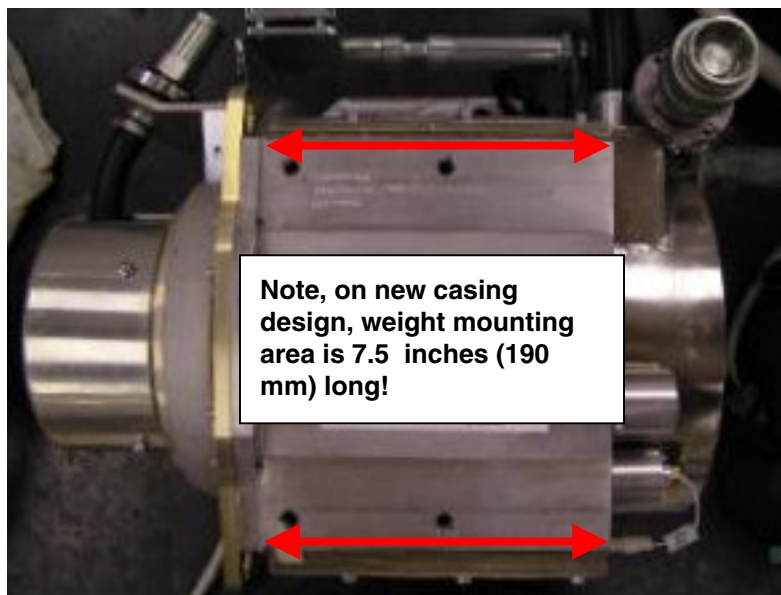
Inspect / measure weight mounting area as indicated in pictures below;

If weight mounting area is ('=') 4.9 inches (125mm) as in first picture. This is the older style casing, screws do not require replacing. Replace covers on gantry, verify system is working properly and close dispatch as Code 2 = FMI NOT REQUIRED PER FMI INSTRUCTION.

If weight mounting area is greater ('>') than 4.9 inches (125mm) as indicated in the second picture below, then proceed to Screw Replacement Procedure on page 7.



Picture above is the old casing design – Screw Replacement NOT REQUIRED.



Picture above is the current casing design – Screw Replacement REQUIRED,

Procedure

This procedure describes the process of replacing VCT/100 tube window screws. For consignment tubes, skip procedure steps 1 – 4.

CAUTION Tube oil expands with the heat generated during exposures and will contract during cooling. The tube should not be pulled off while hot for this procedure.

1. Remove the top, front, and side gantry covers.
2. Position the gantry for tube removal.
3. Use LOTO procedure to make sure that all power to the gantry is off.
4. Proceed with removing the tube from the gantry per the tube replacement section of the system service procedure.

NOTE: As a precaution, use nitrile gloves as hand PPE during component removal, cleaning, and replacement. Avoid any direct contact with the dielectric oil.

5. After protecting the floor with appropriate available material, lower the tube to the floor such that the tube window is up.

IMPORTANT: Protect HV cable when lowering to floor so damage does not occur.

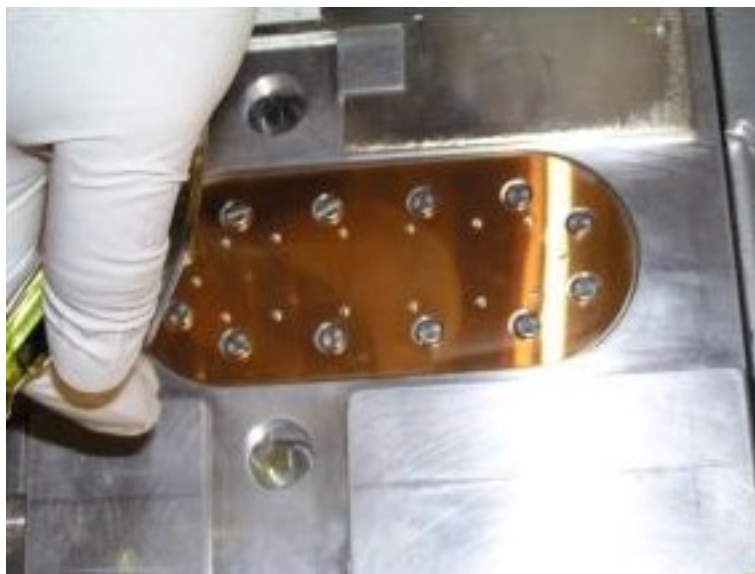
6. Using a small Phillips screwdriver, remove and collect the screws bolting the plated lead window shield to the tube casing aluminum window. New screws have been provided in the kit, so these will be discarded.



7. Remove plated lead window shield from tube.
8. Wipe plated lead window shield with alcohol to clean. The plated lead window shield will be re-installed later. Do not discard.
9. Thoroughly spray the plastic (Ultem) window with alcohol once removed from tube. Use compressed air to blow off all particles. Spray Ultem window with alcohol again, then wipe with lint free cloth. The Ultem window will be re-installed later, **do not discard**.

10. Carefully remove the thin copper filter. Put aside for possible use later.

NOTE: New copper filter is provided for replacement.



11. Inspect the aluminum casing window for traces of oil. Using clean wipes and alcohol, clean any traces of oil at the window and around the screws.



NOTE: In the following steps, each of the 12 screws holding the aluminum window to the casing will be removed one at a time and replaced with the correct screw, provided in the FMI kit. For easy identification, mark all 12 flat screws with a Sharpie so when you remove and replace them they are easy to identify. (Be careful not to erase markings with the cleaning alcohol.)

NOTE: If at this point, there are less than 12 screws, please return to page 6 of this document and review step 3 in **Before You Begin** Section to see if this procedure is required!

12. Mark the head of all 12 screws with a Sharpie to identify the old screws.



13. Using a flat screwdriver, remove **only one** of the twelve flat screws at a time that fix the aluminum window to the casing.

WARNING: This hole is an entrance into the tube. DO NOT allow any substance to get into the tube.



14. Clean with a wipe any burr or debris at countersink surface location. Avoid leaving any of the wipe or cloth within the screw hole.



15. Take a new screw provided in this FMI and hand screw the component until the head bottoms out. To avoid cross-threading the screw or housing, make sure the new screw is perpendicular to the aluminum plate.



16. Use a calibrated torque wrench to tighten and torque the screw to 3 N-m. Once screw is torqued, do not remove, unless replacing.

3 N-m	26.5 lb-in	2.2 lb-ft	30.6 Kg-cm
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17. Repeat Steps 13 to 16 (**ONE SCREW AT A TIME**) with the other screws until all twelve screws are replaced.



NOTE: Replace Nitrile gloves now. The gloves must be clean for the remaining steps.

18. It is important to direct the alcohol into the 14 threaded holes that are visible, around the heads of the screws that were just replaced, and around the outside of the aluminum window. Verify that there are no loose particles.



19. Wipe up all alcohol.
20. Use compressed air to blow remaining oil / alcohol out of 14 threaded holes, around screw heads, and around aluminum window.



21. Repeat steps 18 and 20 as needed until all visible oil is removed. Allow time for alcohol to dry before proceeding to Step 22.

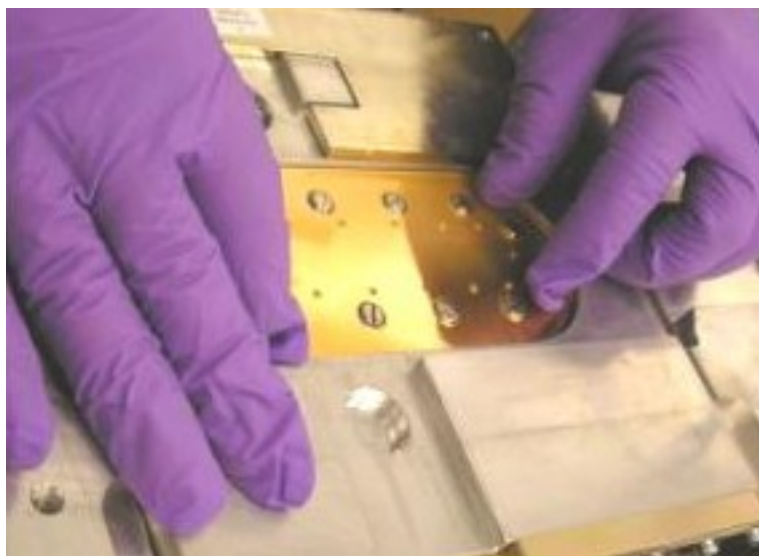
NOTE: Inspect gloves for cleanliness and replace if necessary.

22. Carefully remove new copper filter from bag. There should be no dents or other damage to this filter.

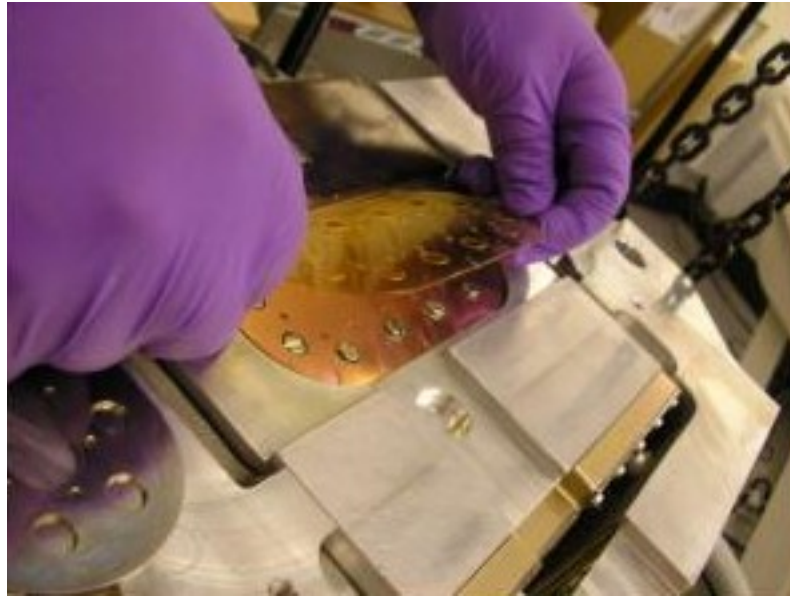
IMPORTANT: In removal from the bag and handling of the copper filter, do not touch the center of the filter. Hold the filter from outside the mounting holes. Do not bend the filter. If there is any damage to the new copper filter, clean and reuse the old one if possible.



23. Place copper filter over aluminum window.



24. Inspect Ultem window for traces of oil. Re-clean if necessary. Install clean Ultem window on top of copper filter.



25. Place the plated lead window shield on top of the Ultem window. Align the screw holes by starting two of the new screws, one on each end of the shield. Hand-tighten only at this time. Install the remaining screws (hand-tight) by alternating screws from side-to-side for an even fit.



26. Using a Phillips screwdriver, tighten the screws in place.

NOTE: Over-tightening the screws can deform the countersink holes at the plated lead window shield. Apply just enough torque to the screw to fix the window in place.

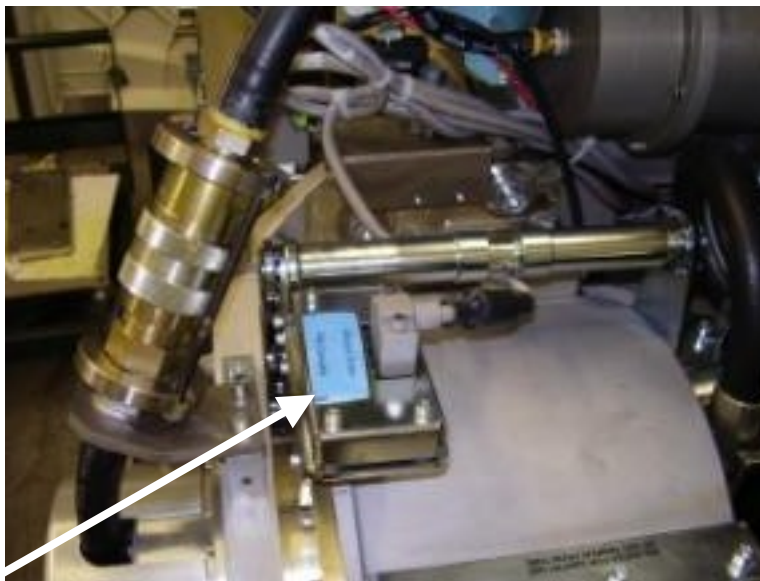


27. The procedure for screw replacement is complete.

Finalization

NOTE: For consignment tubes, skip steps 2 & 4.

1. Apply sticker to smart ID board as shown in illustration below to indicate that the tube has been reworked.



2. Dispose of copper filter and old / extra screws, ensuring that no loose screws are left inside the equipment.
3. Place the tube back in the gantry following the standard service procedure for tube replacement.
4. Perform successful Image Quality testing (Appendix A) to check for artifacts.
5. Close FMI per regional procedure. Remove FMI document from customer site.

Appendix A - Image Quality Testing

Perform the following Image Quality verification checks:

1. Position the QA phantom and level it.
2. Landmark on the line provided and perform phantom centering using the [Scanner Utilities] [Phantom-centering].
3. Depending upon the product, run one of the following series:

VCT	Run the ImgSer 20cmQA VDAS64 series (protocol 45.7) scans by selecting them from the [Service] [Manufacturing] list of protocols. Run all series.
HP 100 and RT 16	Run the ImgSer 20cmQA series (protocol 45.9) scans by selecting them from the [Service] [Manufacturing] list of protocols. Run all series.

4. Using the [Image Analysis] tool, analyze each scan. When using the [Image Analysis] tool, run the analysis in order of the series scanned from the image analysis menu for the phantom used.

NOTE: In the Image select interface on the [Service Desktop], remember to select the top image and drag down to the bottom of the image window holding the left mouse button to select the entire series of images for analysis. This enables the report to show the average values that are compared to specifications and shown with a pass/fail result.